

Do Duc Anh Nguyen

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EXPERIENCE

IMT Atlantique

Doctoral Researcher 01/2023 - present, Rennes, France

Decentralized Reaction Against Malware Propagation 🔄

- Built testbeds to analyze WannaCry traffic and validate propagation models and countermeasures across scales.
- Applied heuristic segmentation to extract abstract malware network-behavior states and design a generic malware model.
- Designed fast decentralized reactions using MirageOS microservices, self-stabilizing vertex cover, and traffic-based signaling.

Network Security Research Intern 03/2022–08/2022, Rennes, France

- Studied and Reproduced an OpenStack/Kubernetes testbed implementing the IETF I2NSF framework.
- Designed and developed a security-policy conflict detection and resolution mechanism for intent-based networking.

Zalo Group

Backend Developer 06/2018 - 06-2020, HCMC, Vietnam

User-Feature Development for the Zalo Application

- Developed backend servers (JSP) supporting user-facing features for the Zalo application.
- Ensured service reliability for over 100 million daily users through testing, optimization, and custom monitoring tools.
- Built user-friendly internal tools (HTML, JavaScript), enabling product teams to interact with databases and feature functions.

ML4U Research Group

Research Member 09/2019 - 10-2019, HCMC, Vietnam

Deep Reinforcement Learning Traffic Light Control (Thesis score: 9.6/10)

- Studied and trained an agent to control traffic light in 3-way and 4-way intersections using Keras, SUMO. 🔄
- Organized a seminar every two weeks.

Teaching & Scientific Reviewing

Network Security (TLS, DNSSEC, IPsec tutorials) Rennes, France
38h, for undergraduate students 2024–2026

Reviewer at IEEE/IFIP NOMS 2025

Publications

- **Nguyen**, Alain, Autrel, Bouabdallah, François, and Doyen. How Fast Does Malware Leveraging EternalBlue Propagate? The Case of WannaCry and NotPetya. SecSoft Workshop, 2024. 📄 📄 📄
- **Nguyen**, Alain, Autrel, Bouabdallah, and François. "Intent-Based Attack Mitigation Through Opportunistic Synchronization of Microservices." NetSoft PhD Symposium, 2024. 📄 📄 📄
- **Nguyen**, Autrel, Bouabdallah, and Doyen. "A Robust Approach for the Detection and Prevention of Conflicts in I2NSF Security Policies." IEEE/IFIP NOMS, 2023. 📄 📄 📄
- **Nguyen**, Autrel, Bouabdallah, Doyen, François, and Alain. "Toward a Network Behavior Model for Self-Propagating Malware: The Case of WannaCry." in submission. 📄

SUMMARY

My main interest is network security engineering. I am motivated by analyzing how attacks implemented to design detection and mitigation mechanisms.

Education

Phd in Cybersecurity 2023-2026

IMT Atlantique Rennes, France

Supervisors: Pierre Alain, Fabien Autrel, Ahmed Bouabdallah, Jérôme François, and Guillaume Doyen

Master in InfoSec 2020-2022

University of Limoges, France

Rank: 5/20

Bac. of Engineering (CS) 2015-2019

HCM University of Technology, Vietnam

Honors program

Honors/Awards

- 2020 Two-year Master's Scholarship
- 2019 Outstanding Student Award
- 2018 Honors Student Award
- 2015 Top-0.1% National Exam Award

Personal Projects

- 2025 Unikernel intrusion detection system detects and embeds reaction in traffic (2 week) 🔄
- 2025 Unikernel firewall apply reaction from embedded traffic (2 week) 🔄
- 2021 Implementation IDS using Random forest (1 week) 🔄
- 2020 Tool applies filter to image using GPGPU (1 week) 🔄

Technical Skill

- ★ Skilled in security research and engineering.
- 🔒 – Malware propagation analysis: traffic inference and modeling
- Development: implementation, evaluation
- Countermeasure: design, analysis
- Network and security-policy analysis
- 🔗 Python, C/C++, Java, OCaml, PowerShell, JavaScript, HTML
- ✂ Linux, OMNeT++, GNS3, Wireshark, Docker, Kubernetes, OpenStack, Git

Languages

English Full professional proficiency
French Elementary / Early B1
Vietnamese Native